Tutorial No.: 9
Engineering Mechanics (ME-101)
Date: 26/03/2024 and Time: 7.55 to 8.55 AM

Question 1: Each of the three balls in Figure 1, has mass m and is welded to the rigid equiangular frame of negligible mass. The assembly rests on a smooth horizontal surface. If a force F is suddenly applied to one bar shown, determine (a) the acceleration of the point O and (b) the angular acceleration $\ddot{\theta}$ of the frame. **[10 marks]**

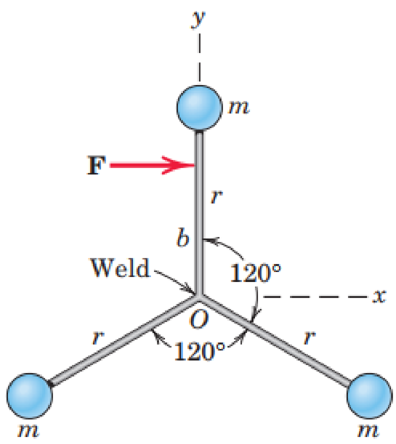


Figure 1

Solution 1: Note, O represents the center of mass of the system. Writing equation of motion in the x direction,

$$(a)\sum F_x = (\sum m_i)a_{Ox} \Rightarrow \; a_{Ox} = \frac{F}{3m} \quad \text{(along positive } x \text{ direction)} \qquad \textbf{[5 marks]}$$

Since, $\sum F_y = 0$, $a_{Oy} = 0$. Now, considering moment of all external forces about center of mass O,

$$(b)\sum M_c = \dot{H}_c \Rightarrow \; bF = 3 \times \frac{d}{dt}(r \times mr\dot{\theta}) \Rightarrow \ddot{\theta} = \frac{bF}{3mr^2} \quad \text{(clockwise)} \qquad \textbf{[5 marks]}$$

Question 2: The small car shown in Figure 2, has a mass of 20 kg, rolls freely on the horizontal track and carries the 5-kg sphere mounted on the light rotating rod with $r = 0.4$ m. A geared motor drive maintains a constant angular speed $\theta = 4$ rad/s of the rod. If the car has a velocity $v = 0.6$ m/s when $\theta = 0°$, calculate v when $\theta = 60°$. Neglect the mass of the wheels and any friction. **[10 marks]**

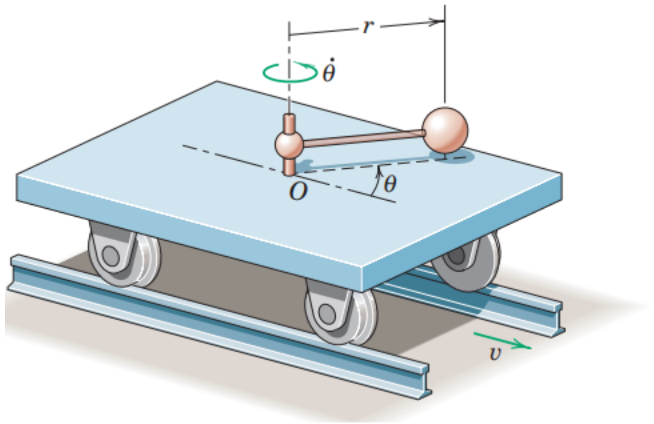
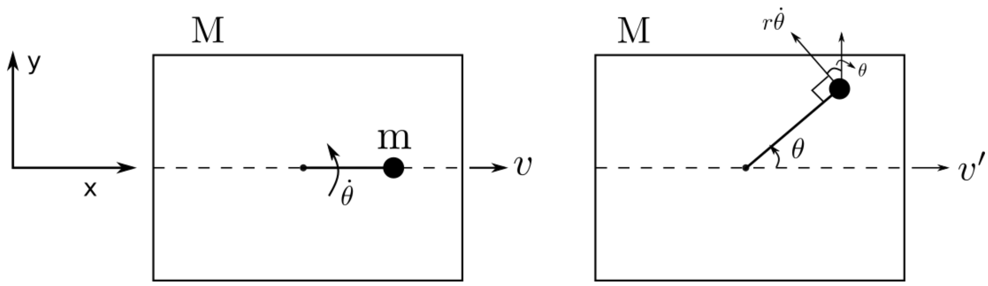


Figure 2

Solution 2:



Since there are no external forces on the cart and the ball assembly, thus its linear momentum in the x-direction would be conserved.

$$G_{2x} = G_{1x}$$

$$Mv' + m(v' - r\dot{\theta}\sin\theta) = (M+m)v \qquad \textbf{[6 marks]}$$

$$v' = \frac{(M+m)v + mr\dot{\theta}\sin\theta}{M+m} = 0.877 \text{ m/s} \qquad \textbf{[4 marks]}$$

Can we apply linear momentum conservation along y-direction?
$\Rightarrow$ **No**, because of the reaction forces on the wheel in the y-direction.

Question 3: The two spheres, depicted in Figure 3, are rigidly connected to the rod of negligible mass and are initially at rest on the smooth horizontal surface. A force F is suddenly applied to one sphere in the y-direction and imparts an impulse of 10 N-s during a negligibly short period of time. As the spheres pass the dashed position, calculate the velocity of each one.

[10 marks]

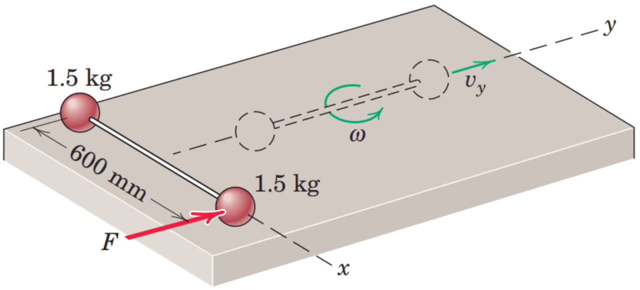
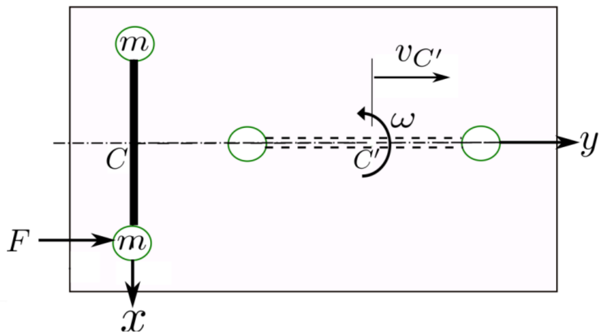
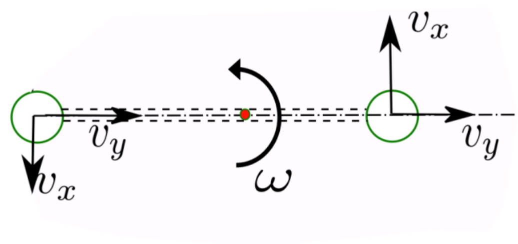


Figure 3

Solution 3:





Using linear-impulse-momentum relation

$$\int \Sigma F_y dt = G_{2y} - G_{1y}, \tag{1}$$

$$\implies 10 = 2(mv_{C'}) - 0, \qquad \textbf{[2 marks]}$$

$$\implies v_{C'} = \frac{5}{1.5} = \frac{10}{3} \text{ m/s.} \qquad \textbf{[2 marks]} \tag{2}$$

Invoking angular impulse-angular momentum relation about centre of mass,

$$\int \Sigma M_C dt = H_{C'} - H_C.$$

$$\implies \int F\left(\frac{l}{2}\right)dt = 2 \times \underbrace{\left(\frac{l}{2} \times mv_x\right)}_{C.C.W} - 0. \qquad \textbf{[2 marks]}$$

$$\implies v_x = \frac{10}{3} \text{ m/s.} \qquad \textbf{[2 marks]}$$

Note, $v_y = v_{C'}$. Thus, the magnitude of velocity of both the balls is,

$$|\bar{v}_b| = \sqrt{v_x^2 + v_y^2} = \frac{10\sqrt{2}}{3} \text{ m/s} = 4.71 \text{ m/s.} \qquad \textbf{[2 marks]}$$

Note: Angular velocity of the assembly can be obtained as $\omega = \frac{v_x}{l/2} = \frac{100}{3}$ rad/s.

Question 4: The three small spheres, each of mass m, are secured to the light rods to form a rigid unit supported in the vertical plane by the smooth circular surface (refer Figure 4). The force of constant magnitude P is applied perpendicular to one rod at its midpoint. If the unit starts from rest at θ =0, determine (a) the minimum force P_{min} which will bring the unit to rest at $\theta = 60^0$, and (b) the common velocity v of spheres 1 and 2, when $\theta = 60^0$, if $P = 2P_{min}$.

[10 marks]

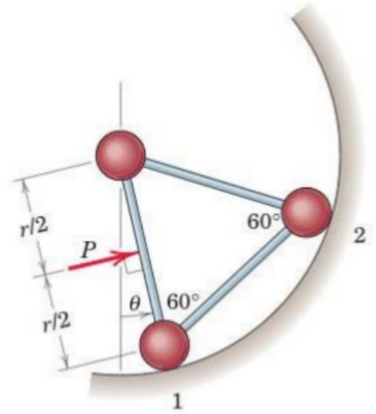
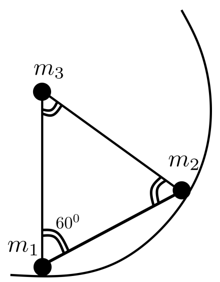


Figure 4

Solution 4:



$\theta = 0$

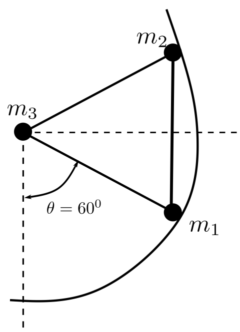
(a)

$$W_{1-2} = V_2 - V_1 = \Delta V$$

$$\implies P_{min}\left(\frac{r}{2} \times \frac{\pi}{3}\right) = m_1 g(r - r\cos\frac{\pi}{3}) + m_2 gr \qquad \textbf{[3 marks]}$$

$$= \frac{3}{2}mgr \qquad .....(m_1 = m_2 = m)$$

$$P_{min} = \frac{\cancel{6}^{\,3}}{\pi \cancel{r}} mg\cancel{r}\frac{3}{\cancel{2}} = \frac{9mg}{\pi} \qquad \textbf{[2 marks]}$$



(b)

$$2P_{min}\left(\frac{r}{2} \times \frac{\pi}{3}\right) = \Delta V + \Delta T \qquad \textbf{[2 marks]}$$

$$\text{now, } \Delta V = P_{min}\left(\frac{r}{2} \times \frac{\pi}{3}\right)$$

$$\implies \Delta T = P_{min}\left(\frac{r}{2} \times \frac{\pi}{3}\right)$$

$$2 \times \frac{1}{2}mv^2 = \frac{9mg}{\pi} \times \frac{\pi r}{6} \qquad \textbf{[2 marks]}$$

$$\implies v = \sqrt{\frac{3gr}{2}} \qquad \textbf{[1 mark]}$$

Question 5: As shown in Figure 5, a high speed jet of air issues from the nozzle A with a velocity of v_A and mass flow rate of 0.36 kg/s. The air impinges on a vane causing it to rotate to the position shown. The vane has a mass of 6 kg. Knowing that the magnitude of the air velocity is equal at A and B, determine, (a) the magnitude of the velocity at A, and (b) the components of the reactions at O. **[10 marks]**

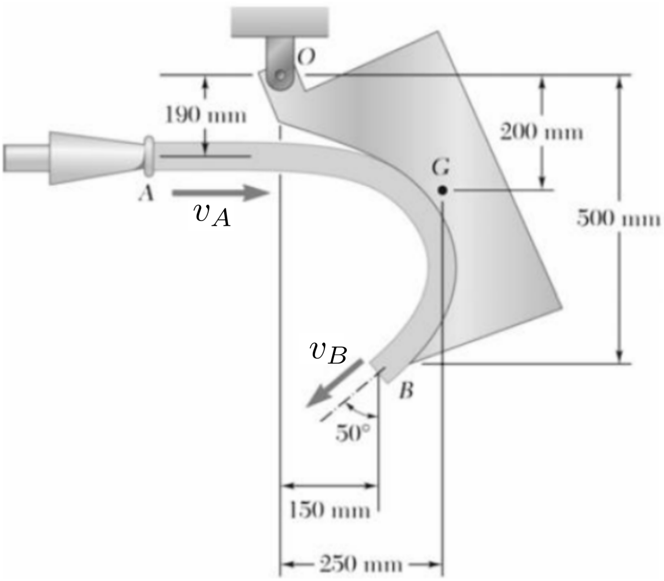
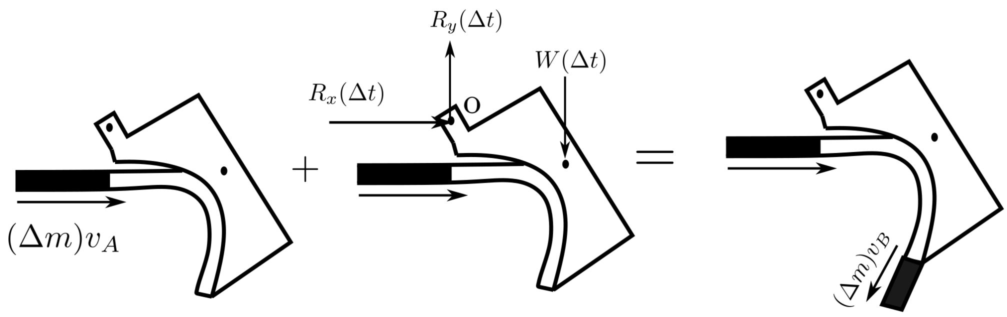


Figure 5

Solution 5: Assume that the speed of the air jet is same at A and B. i.e, $v_A = v_B = v$
Apply the principle of impulse and momentum.



(a) Using angular-impulse-momentum relation about O (+ ↶):

$$(0.190)(\Delta m)v - (0.250)W(\Delta t) = -(0.150)(\Delta m)v\cos 50^0 - (0.500)(\Delta m)v\sin 50^0 \quad \textbf{[2 marks]}$$

$$\begin{aligned} v &= \frac{\Delta t}{\Delta m} \cdot \frac{0.250W}{0.150\cos 50^0 + 0.500\sin 50^0 + 0.190} \\ &= \frac{0.250W}{0.669}\left(\frac{1}{\dot{m}}\right) \\ &= \frac{(0.250)(6)(9.81)}{(0.669)(0.36)} \implies v = 61.058 \text{ m/s} \end{aligned} \quad \textbf{[1 mark]}$$

(b) Using linear-impulse-momentum relation along x:

$$(\Delta m)v + R_x(\Delta t) = -(\Delta m)v\sin 50^0 \qquad \textbf{[2 marks]}$$

$$R_x = -\frac{\Delta m}{\Delta t}v(1+\sin 50^0)$$

$$= -(0.36)(61.058)(1+\sin 50^0)$$

$$\implies R_x = -38.82 \text{ N} \qquad \textbf{[1 mark]}$$

(c) Using linear-impulse-momentum relation along y:

$$0 + R_y(\Delta t) - W(\Delta t) = -(\Delta m)v\cos 50^0 \qquad \textbf{[2 marks]}$$

$$R_y = W - \frac{\Delta m}{\Delta t}v\cos 50^0$$

$$= (6)(9.81) - (0.36)(61.058)\cos 50^0$$

$$\implies R_y = 44.73 \qquad \textbf{[1 mark]}$$

$$R = \sqrt{(38.82)^2 + (44.73)^2} = 59.2 \text{ N}$$

$$\tan\alpha = \frac{44.73}{38.82} \implies \alpha = 49^0 \qquad \textbf{[1 mark]}$$

Question 6: The end of a chain of length L and mass ρ per unit length which is piled on a platform is lifted vertically with a constant velocity v by a variable force P (refer Figure 6). Find P as a function of the height x of the end above the platform. Also find the energy lost during the lifting of the chain. **[10 marks]**

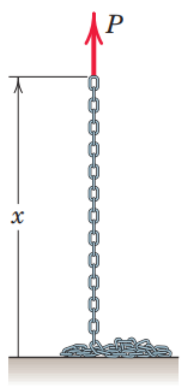
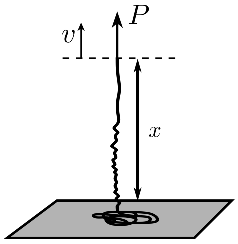


Figure 6

Solution 6:



In case of variable mass system we have,

$$\sum F = m\cancelto{0}{\dot{v}} + \dot{m}u \qquad ....(v = \text{ constant}) \qquad (3)$$

[1 mark]

Here, the rate of mass intake, $\dot{m} = \rho\dot{x} = \rho v$

The magnitude of the relative velocity of the mass stream,

$u = v - v_0 = v$

Now, using Equation 3

$$\sum F = P - \rho x g = (\rho v)v$$

$$\implies P = \rho(xg + v^2)$$

[4 marks]

The work done in lifting the whole chain is,

$$W_{1-2} = \int_0^L P dx = \frac{1}{2}\rho g L^2 + \rho v^2 L$$

[2 marks]

Total change in energy of the system,

$$\Delta T + \Delta V = \left[\frac{1}{2}(\rho L)v^2 - 0\right] + \left[(\rho L)g(\frac{L}{2}) - 0\right]$$

[2 marks]

Hence, the total energy loss in lifting the whole chain,

$$\Delta E = W_{1-2} - (\Delta T + \Delta V) = \frac{1}{2}\rho L v^2$$

[1 mark]